

GENERAL SPECIFICATION G7.7

Telephone System

CONTENTS

G7.7 Telephone System.....1

 G7.7.1 General.....1

 G7.7.2 Telephone Cabling1

 G7.7.3 Cable Laying2

 G7.7.4 Connector Termination - Copper Cable4

 G7.7.5 Testing.....8

 G7.7.6 Telephone System Functionality.....8

G7.7 TELEPHONE SYSTEM

G7.7.1 General

- (a) Where the Particular Specification requires telephone lines for remote dial, voice and data circuits in facilities, the Contractor shall be responsible for initiating the liaison with the local telecommunications provider with sufficient lead time given to the Board on when the lines are to be put into service.
- (b) The demarcation point shall be at the Distribution Point (DP) termination panel provided by the local telecommunications provider. All connection, installation and other charges from the DP to the required communication equipment shall be borne by the Contractor. The required communication equipment shall include those proposed by the Contractor, as specified elsewhere in the Contract as well as those supplied by the local telecommunications providers.
- (c) Any security deposit required by the telecommunications provider with regard to the operation of credit facilities for telecommunications services shall be paid by the Board and held in the name of the Board.
- (d) The Contractor shall indemnify the Board with respect to any call and usage charges made up until the start of the Defects Liability Period (DLP).
- (e) The Contractor shall comply with the licensing and the regulations as defined by IMDA (Info-communications Media Development Authority) and as defined in the latest Code of Practice for Info-Communication Facilities in Buildings (COPIF).

G7.7.2 Telephone Cabling

- (a) Telephone cables for the incoming lines shall have 50% spare cores of the telephone lines required.
- (b) Digital and Voice over Internet Protocol (VoIP) telephone systems shall use an appropriate combination of fibre optic (F/O) and CAT6 cabling. Cables shall comply with General Specification G6.2 - Installation Requirements.
- (c) Analogue telephone system cables shall be UTP cable to appropriate EIA standards. The conductors shall be installed with colour coded PVC and jacketed with low friction PVC.
- (d) Analogue telephone system cabling for the interconnection of indoor telephone terminal cabinets shall consist of multiple twisted pairs of a minimum 0.5mm (solid) annealed copper conductors with PVC insulation and minimum of 25 pairs.
- (e) Analogue telephone system cables run underground shall be in ductwork/conduits and shall consist of twisted pairs of a minimum 0.65mm annealed copper conductors with colour coded PVC insulation, surrounded by a wrapped and coated aluminium shield. The cable shall be jacketed with a high molecular weight polyethylene and shall be flooded with a high dielectric waterproofing compound and shall be steel armoured and gel-filled.

G7.7.3 Cable Laying

- (a) Containment
 - (i) Internal cables shall be laid in cable containment systems, trunking, trays or conduit.
- (b) Separation from Power Cables
 - (i) At no point shall communications cables be tied to power cables or run along the same conduits or ducts or run through the same connection box.
 - (ii) Internal cables run in metal enclosed trunking or metal conduit shall be separated from external parallel runs of power cables by a minimum of 50mm. Refer also to General Specification G6 – ICA.
 - (iii) Cable runs within desks or consoles shall have power and communications cables separated, as described in BS 6396.
- (c) General Cabling Principles
 - (i) Cables shall be fixed in straight parallel runs and shall be held in place by broad cable ties at not more than 600mm centres on horizontal trays and 300mm on vertical trays.
 - (ii) Cables shall not be held so tightly or by sufficiently narrow cable ties that permanent indentation (sufficient to measurably affect the transmission characteristics of the cable) or cutting of the outer sheath or conductor insulation occurs.
 - (iii) The manufacturers minimum bending radii requirements shall be adhered to during cable installation.
 - (iv) There is no requirement for tray bridges to separate voice and data cables. Voice and data cables shall cross over each other at right angles.
 - (v) The maximum unsupported communications cable length shall be 100mm. Suitable Unistrut, tray supports and tray bends shall be installed where required. All support steelwork shall be suitable earthed.
 - (vi) Installed cable shall take the shortest route consistent with the usage of the containment system and defined routes, and shall not, unless specified, contain length in excess of the distance required.
 - (vii) All cables, whether on trays, in ducts or laid directly on the concrete floor, which cross service pipes carrying hot or refrigerant fluids shall be protected from excessive high or low temperatures or condensation. This may require separation and/or thermal insulation.
 - (viii) Cables shall be kept clear of fire suppression outlets, heat and smoke detectors and water detector strips.
 - (ix) Installed cables shall not be laid where they may be subject to vibration or movement.
 - (x) Cables shall not be fixed using staples.

- (xi) The metal foil braid or screen of FTP cables shall be connected to earth at both ends of the cable, provided that these are measured to be within 1V potential of each other. If this potential is exceeded, the cable Contractor shall not install the cable but shall advise the Main Contractor.
- (d) Cables on Trays
 - (i) Voice and data cables may be run on the same tray. In such cases, voice and data cables shall be kept to opposite sides of trays and shall be separated by a minimum of 50mm, where practical.
 - (ii) Cable bundles shall not run over hard or sharp edges such as the edges of trays. Particular care shall be taken that cables are not laid on tray bridges, or other features, that present sharp or small radius edges to the cables.
 - (iii) Cables shall leave trays at right angles and shall be suitably cable tied and dressed.
 - (iv) Where basket tray is installed, cables shall exit through the bottom or side of the tray.
 - (v) Cable exits shall be strictly controlled such that additional cables can be easily and neatly added to trays in the future.
 - (vi) Cables shall be installed to use the minimum of the tray width.
 - (vii) Cables shall not be stacked higher than 50mm on basket trays at any location. Cables shall not be stacked higher than 150mm on smooth surfaces. The manufacturer's recommendations are not to be exceeded.
- (e) Cables in Trunking
 - (i) Voice and data cables shall be run in separate compartments of trunking, where practical.
 - (ii) Trunking compartments shall be no more than 50% full to allow for future installation of cables.
 - (iii) Cables shall enter and exit trunking via bridges to ensure complete separation between communications and power compartments.
- (f) Cables in Conduit
 - (i) The number of cables in each conduit shall be controlled to allow for future cable installation and to stay within the manufacturers maximum allowable cable pulling tension.
 - (ii) All draw wires shall be replaced during each cable pull.
- (g) External Cables
 - (i) External cable shall be run in underground ducts. The Cable Contractor shall prove that they are clear prior to pulling of cables. The Cable Contractor shall satisfy himself that the cable can be installed without excessive force on the cable.
 - (ii) Cable pulling tension shall not exceed manufacturers limits. The cable pulling tension shall be applied smoothly without jerks and at no time shall it exceed the manufacturer's limits.

- (iii) The pulling equipment shall be arranged either to indicate the pulling tension, which shall be continually monitored during a pull, or shall mechanically assure that the maximum permissible tension is not exceeded.
- (iv) Curvature of cables shall not exceed that permitted by the manufacturer.
- (h) External - Internal Joints
 - (i) Internal grade cables shall be jointed to external grade cables within 3 metres of the point of duct entries.
 - (ii) Joints shall be carried out in wall-mounted or tray mounted joint boxes. Joint boxes shall be earthed to protective earths either locally or earth cable shall be run back to the nearest protective earth bar.
 - (iii) Aluminium tape or foil screens of external cables shall be earthed at one end only.

G7.7.4 Connector Termination - Copper Cable

- (a) Backboxes
 - (i) Ensure physical compatibility between connector faceplates, provided under this contract.
 - (ii) Pay particular attention to the depth of the connector and clearance within the associated backbox.
 - (iii) Ensure that cables are neatly tied and dressed inside backboxes.
 - (iv) Allow sufficient slack cable on each connector to enable the faceplate to be unscrewed and removed for visual examination or repair, and to allow replacement of connectors at a future date.
 - (v) Ensure that cables within backboxes do not to exceed the maximum permitted curvature.
 - (vi) Install appropriate protective glands, grommets, boots and sleeves, securely fixed at entry points to backboxes.
 - (vii) Ensure that cables are secured or tied to ensure that strain is not transferred to electrical contacts, and that inspection of connectors imposes no movement or strain on individual cable pairs and their terminations. Ensure that strain relief does not damage cable jackets or conductors.
 - (viii) Ensure that strain relief is inherent within connectors or separately mounted.
- (b) Faceplate / Wallboxes
 - (i) Securely fix connectors to faceplates mounted in wall or surface mounted backboxes.
 - (ii) Ensure faceplates are suitable for single or double gang backboxes, and are of white PVC or metal construction
- (c) Patch Panels
 - (i) Provide Copper cable patch panels suitable for mounting in 19" frames within termination cabinets and fixed using securing bolts and captive nuts at either side.

- (ii) Ensure that panels are front mount only, no more than 400mm front to back, and contain return flanges to provide rigidity.
- (iii) Firmly fix snap in connectors to the connector cut-out in the panel. Ensure that the connector layout follows a consistent pattern to the maximum possible density consistent with ease of installation and maintenance.
- (iv) Install RJ45 connectors with plug locking mechanisms at the top.
- (v) Provide sufficient finger space around connectors to allow patch cables to be connected and disconnected and to allow individual connectors to be mounted and dismantled without disturbing other adjacent connectors.
- (vi) Provide panels to accommodate full labelling of every connector as described in the labelling section of this document.
- (vii) Ensure that the termination sequence is consistent on all frames with the lowest number at the first available termination at the top left of the frame or panel, or allocated portion of a panel.
- (viii) Ensure that panels contain cable support at the rear for cable dressing individual cables.
- (ix) Ensure that IDC connectors have a logical layout and are readily accessible from the rear of the panels.
- (x) Ensure that cables terminating in connectors are dressed in the panel with due consideration to minimum bending radii, and that the cable dressing does not cause permanent indentation of cable sheath.
- (xi) Install sufficient cable reserve such that re-termination of connectors can occur in addition to an allowance for removal and inspection of the complete panels.
- (xii) Protect cables entering panels by appropriate protective glands or grommets. Where practical, the external sheath of the cable will not be stripped back outside of the strain relief of the patch panel.
- (xiii) Ensure that enclosed panels are sealed to IP54.
- (xiv) Provide strain relief at each termination such that any strain on the cable is not transferred to the electrical contacts in the connector.
- (xv) Mount panels at the top of each cabinet such that network equipment can be mounted below. Ensure cabinets are not to be more than 50% full of panels.
- (xvi) Separate each set of 2 panels from a neighbouring set of 2 panels by a cable management strip containing 4 jumper rings each capable of supporting up to 100 patch cables.
- (xvii) Provide vertical jumper rings as described in the cabinet specification section.
- (xviii) Provide wall panels housing terminations with maintenance access to rear using sliding, tilting or other mechanism that does not strain the terminated cables or terminations.
- (xix) Earth metal panels to the metal structure of the cabinet and earth the metal structure to building safety earth.

- (d) Installation of cabinets and enclosures
 - (i) Ensure that cabinets are free-standing and bolted together but not be fixed to the floor.
 - (ii) Ensure that patching connections is possible between cabinets via one number 300mm tray installed on the cabinet base and running full length of all cabinet runs.
 - (iii) Connect each cabinet to safety earth by a minimum of 10mm² single core green/yellow PVC earth cable. Provide each cabinet with 1 number earth cable and ensure that cabinet connections are not be daisy-chained.
 - (iv) Provide wall mounted Krone 108A frames and wall mounted RJ45 patch panels fitted with robust lockable covers.
- (e) Identification of cables and connectors
 - (i) Provide full details of the cable label scheme for every cable in the works and obtain the agreement of the Engineer before creating any labels. Cable labels not agreed fully in this way may be subsequently rejected.
 - (ii) Labels shall use an agreed typeface size and font. Hand-written labels will not be accepted, except on a temporary basis during installation, and will not be acceptable for cable testing purposes.
 - (iii) All labels shall be permanently fixed, legible, durable and robust. All labels shall be fixed horizontally on fixed equipment or longitudinally along the line of a cable.
- (f) RJ45 Sockets
 - (i) Provide RJ45 sockets certified to conform to the relevant Category of the EIA/TIA 568 standard.
 - (ii) Ensure that all sockets are sourced from the same manufacturer.
 - (iii) Ensure that RJ45 sockets contain sliding shutters to keep out dust except in equipment room panels.
 - (iv) Ensure that RJ45 sockets contain a locking feature that ensures secure connection of RJ45 patch or drop cables plugged into it.
 - (v) Ensure that RJ45 sockets have the capability to accommodate a label on the connector socket or faceplate.
 - (vi) Ensure that sockets meet or are better than the performance detailed in EIA/TIA 568 and EIA/TIA TSB-40.
 - (vii) Provide cable connection to sockets to conform to the EIA/TIA 568, TSB36 and TSB 40.
 - (viii) Ensure that the twist (and twist rate) in each pair of conductors is maintained as close as possible to the IDC connector.
 - (ix) After a conductor has been terminated, cut excess free conductor length as closely as practicable to the IDC connection block.
 - (x) For FTP, ensure that the metal foil, tape or braid shield is electrically continuous and insulated from each conductor cores and earthed at one end only.

- (xi) Directly terminate the cable conductors at each end on RJ45, 8 pin keyed sockets as defined by EIA/TIA 568 designation T568B.
- (g) Cabinets and enclosures
 - (i) Provide cabinets or enclosures meeting the following specification:
 - 1) Fabricated from steel or aluminium;
 - 2) Two number 19" mounting frames to facilitate shelf and equipment mounting, one frame to be positioned 200mm from the front of the cabinet. The rear frame to be 450mm from the front frame;
 - 3) Minimum usable frame height to be 45U (1U=1.75 inches);
 - 4) Overall cabinet height to be less than 2150mm;
 - 5) Each cabinet to be 600mm wide and at least 800mm deep (Internal measurements);
 - 6) A removable gland plate at least 600mm x 300mm to be provided in the cabinet floor at the rear. The gland plate to have a number of rectangular slots and blanking plates. The rectangular slots to have no sharp edges and shall be fitted with brushed strip;
 - 7) Cabinets to be fitted with heat extract fans in the roof;
 - 8) Front and rear doors to be lockable. All cabinet keys to be identical;
 - 9) A smoked glass front door to be provided;
 - 10) The rear door to have a documentation wallet;
 - 11) Side panels to be removable;
 - 12) One number 150mm tray to be installed up each side of the cabinet at the front with patch cord jumper rings at not less than 300mm intervals. Each patch cord jumper ring to be capable of carrying 200 UTP 4 pair patch cords;
 - 13) One number 300mm tray up each side of the cabinet between the front and rear mounting frames;
 - 14) One number 100mm tray up each side of the cabinet at the rear;
 - 15) Evertically at the rear;
 - 16) 4 mounting feet to be provided for each cabinet; and
 - 17) 4 equipment shelves with front and back supports to be supplied for each cabinet.
- (h) UTP and FTP Patch Cables
 - (i) Provide patch cables tested and certified by the named manufacturer to conform to the relevant Category of EIA/TIA 568;
 - (ii) Provide patch cables to meet the performance characteristics detailed in EIA/TIA 568, EIA/TIA TSB 36 and EIA/TIA TSB 40;
 - (iii) Do not assemble patch cables on site;
 - (iv) Provide patch cables manufactured in fixed lengths;

- (v) Provide patch cables complete with the relevant Category of plugs or connectors at both ends;
- (vi) Ensure that patch cables are "straight through" and do not introduce crossovers of connections; and
- (vii) Ensure that the minimum bend radius of patch and drop cables is 50mm or better.

G7.7.5 Testing

- (a) During the testing procedure, The Cable Contractor shall confirm that each cable and outlet has the correct permanent label. If the label is not correct, the formal test of that cable shall be deferred.
- (b) The purpose of testing is to demonstrate that the workmanship on site has been adequate and it should be noted that a successful test of a cable does not assure that the cable is free from faults. Each cable and termination should be visually inspected for faults and for workmanship standards. Where testing is carried out, the cable test shall demonstrate that the cable installation is to the required standard.
- (c) Cables that fail tests shall be pulled out and replaced by complete runs of new cable. Cable jointing or running of part segments will not be acceptable.

G7.7.6 Telephone System Functionality

- (a) Telephone system functionality shall be as per the Particular Specification.

END OF SECTION